SYRIATEL CHURN PREDICTION.

PREDICTION FOR CUSTOMER RETENTION STRATEGY

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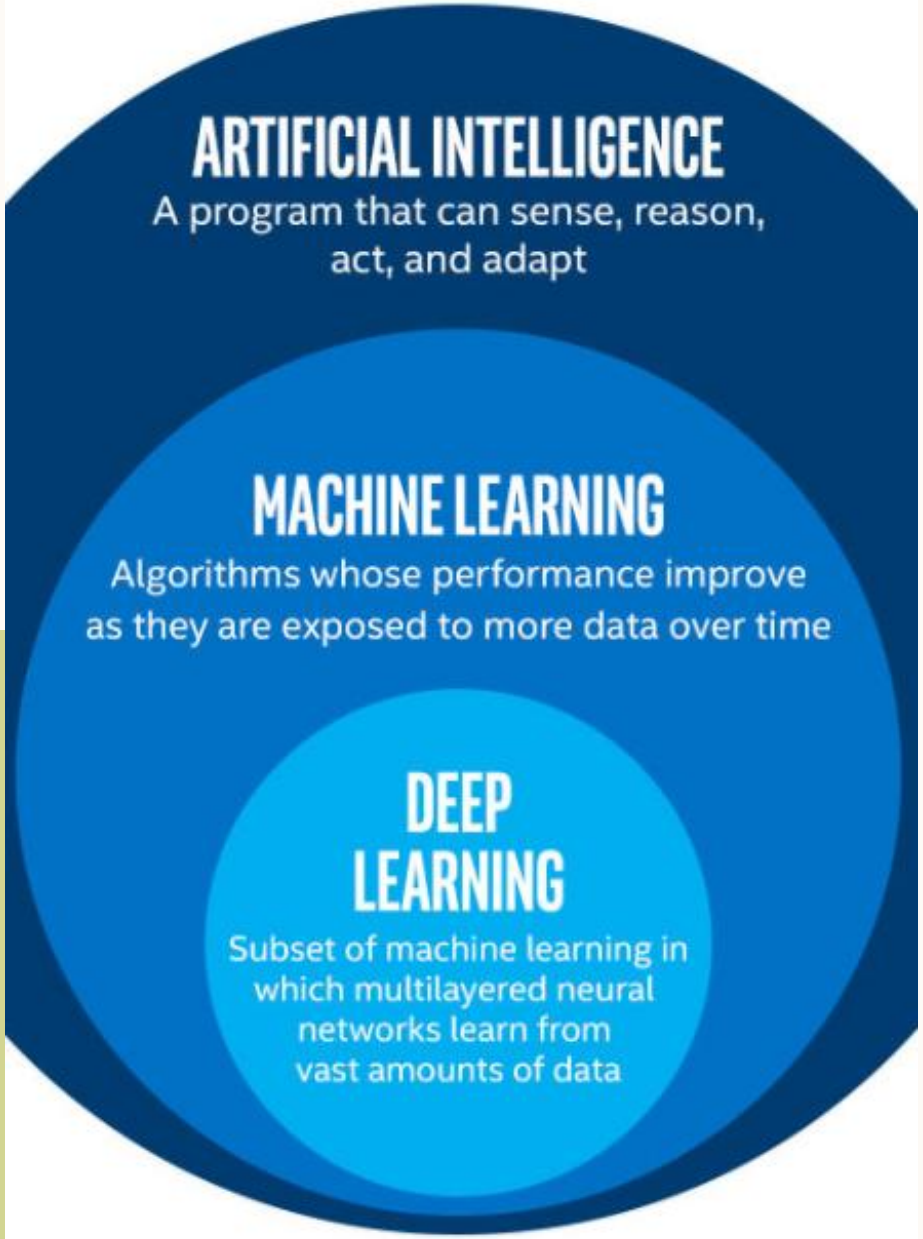
CUSTOMER CHURN

- As per Accenture 2024 research, 77% of consumers are shifting from one telecommunication to another much faster than they did three years ago.
- This means that telecommunication companies (telcos) must put in extra effort to retain their customers.
- What can business do to retain customers ?
- I try to answer the question through Prediction models.



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OBJECTIVE

- Develop a high classification model to identify at risk customers early.
- Enable Syriatel to implement targeted retention measures and stabilize revenue shares.

THE BUSINESS PROBLEM
ATTRITION CRISES

- Rising attrition rates pose a direct threat to revenue stability and competitive market position in the Syriatel landscape
- Maintain profitability in a saturated market
- Identify key drivers of customer defection
- Implement proactive loyalty programs
- Enhance competitive market share

DATASET PROFILE

- 3333 Records – Comprehensive customer activity logs from Kaggle's dataset
- 21 Features- Usage metrics account metadata and service interaction history.
- Customer activity- Data spans call minutes, charges across different times, and plan subscriptions.
- Clean data- Zero missing or duplicate values found during cleaning

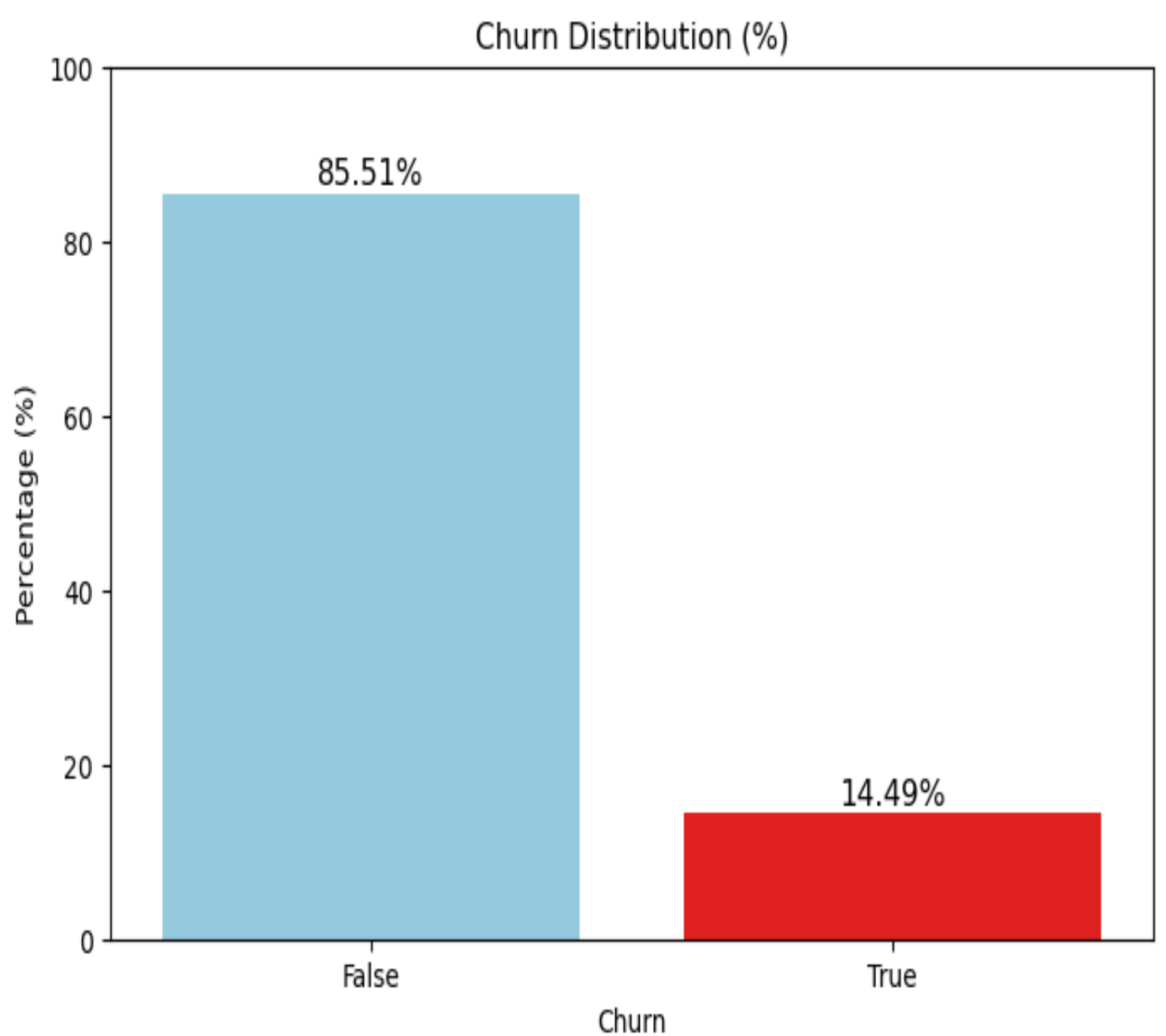
EXPLORATORY DATA ANALYSIS

14.49 % of the 3,333 customers have terminated their contract with Syriatel company.

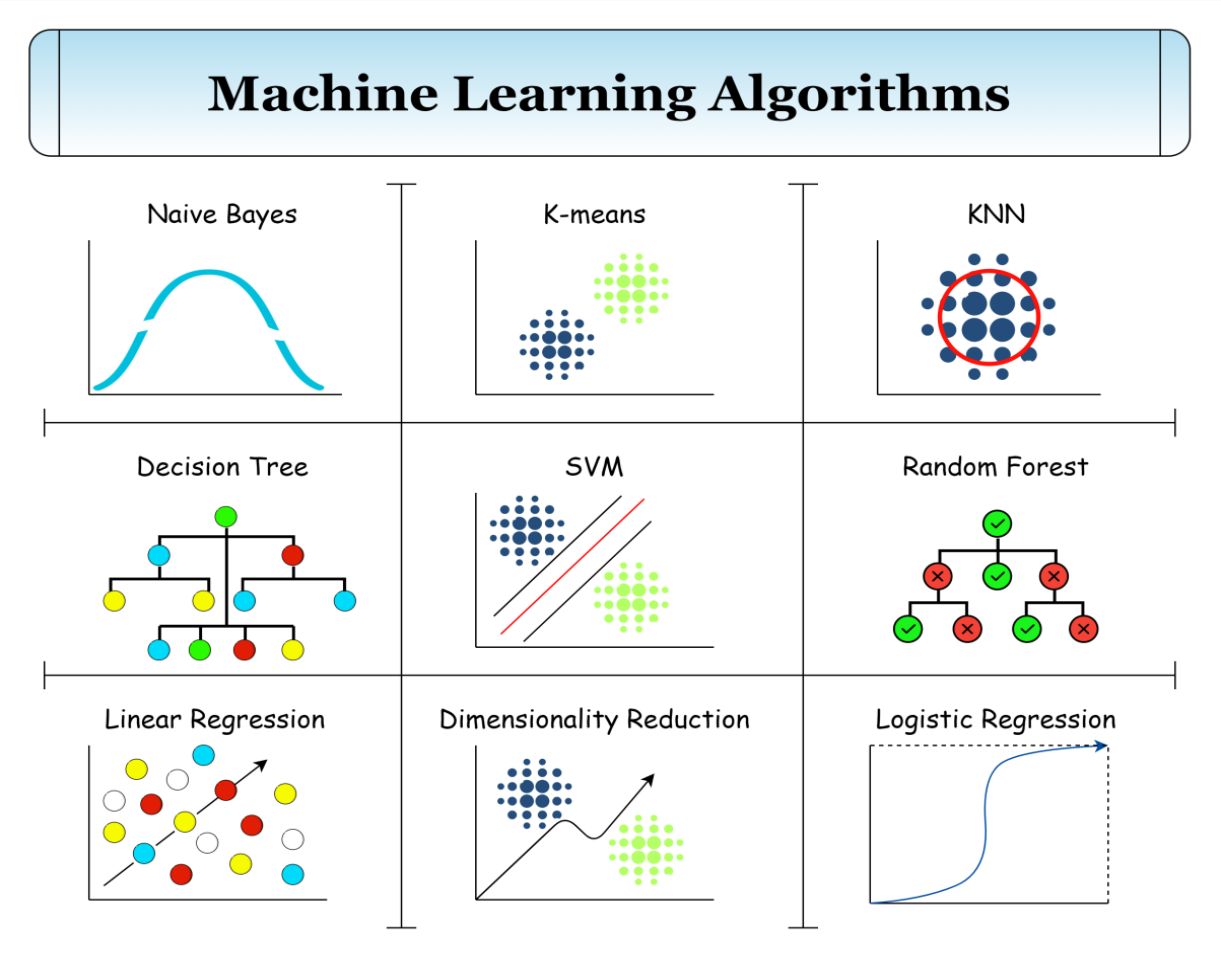
This is a total of 483 customers lost.

False = Not churned

True = Churned



MODEL SELECTION AND EVALUATION



- Logistic Regression- Baseline model for probabilistic classification
- Decision Tree- Non-linear logic capturing complex feature interactions
- Random Forest- Ensemble method reducing variance and overfitting

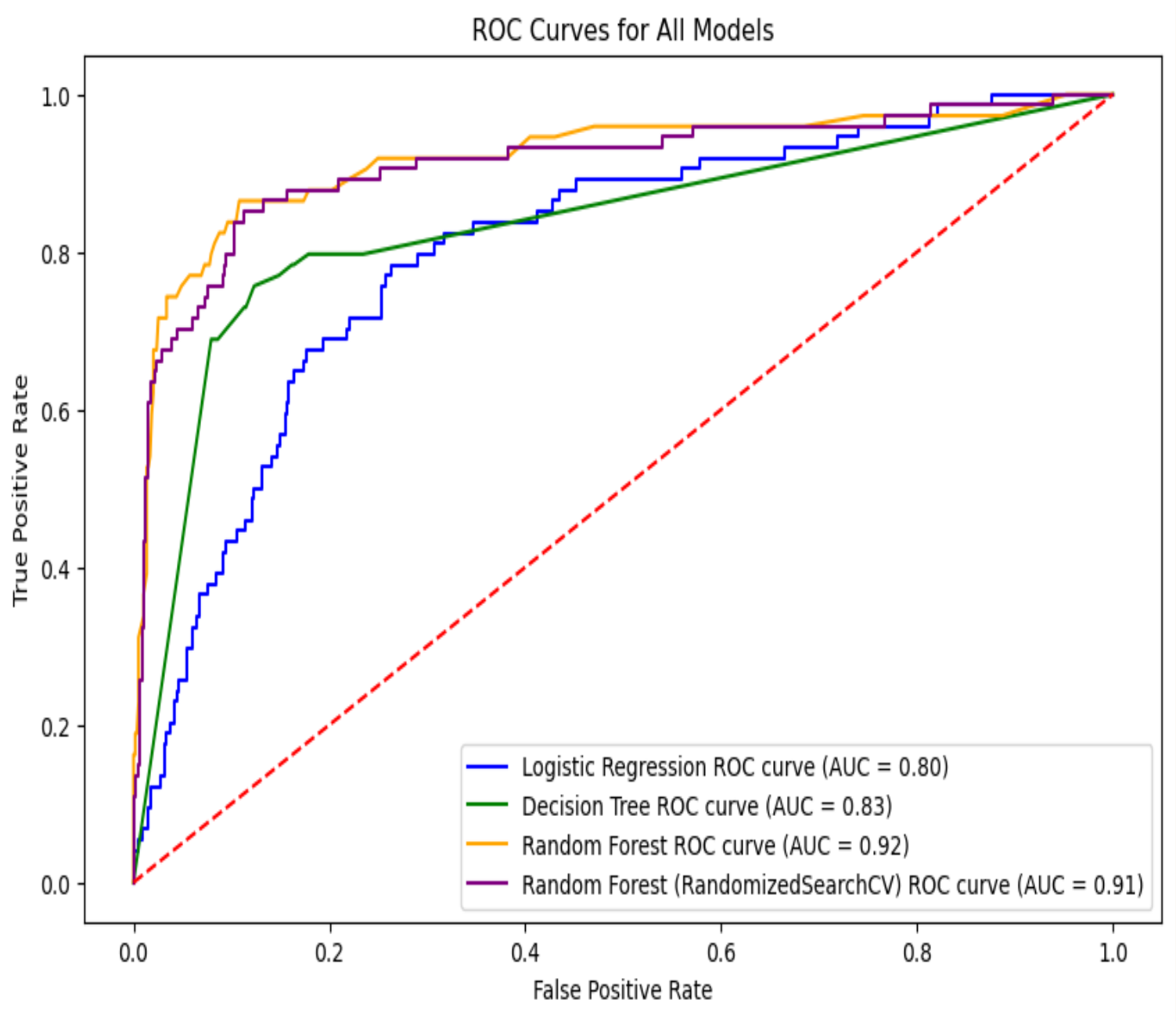
MODEL PERFORMANCE METRICS

Model	Accuracy (Test Set)	F1 Score (Test Set)	Recall (Test Set)	Precision (Test Set)
Logistic Regression	0.91	0.29	0.22	0.4
Decision Trees Classifier	0.89	0.63	0.82	0.5
Random Forest Classifier	0.91	0.66	0.77	0.58

RECEIVER OPERATING CHARACTERISTIC(ROC) AND AREA UNDER CURVE(AUC)

Random Forest Classifier has high ROC Curve approaching to left corner and a high AUC of 0.92,

The Classifier therefore able to distinguish churn and non-churn customers.



MODEL COMPARISON

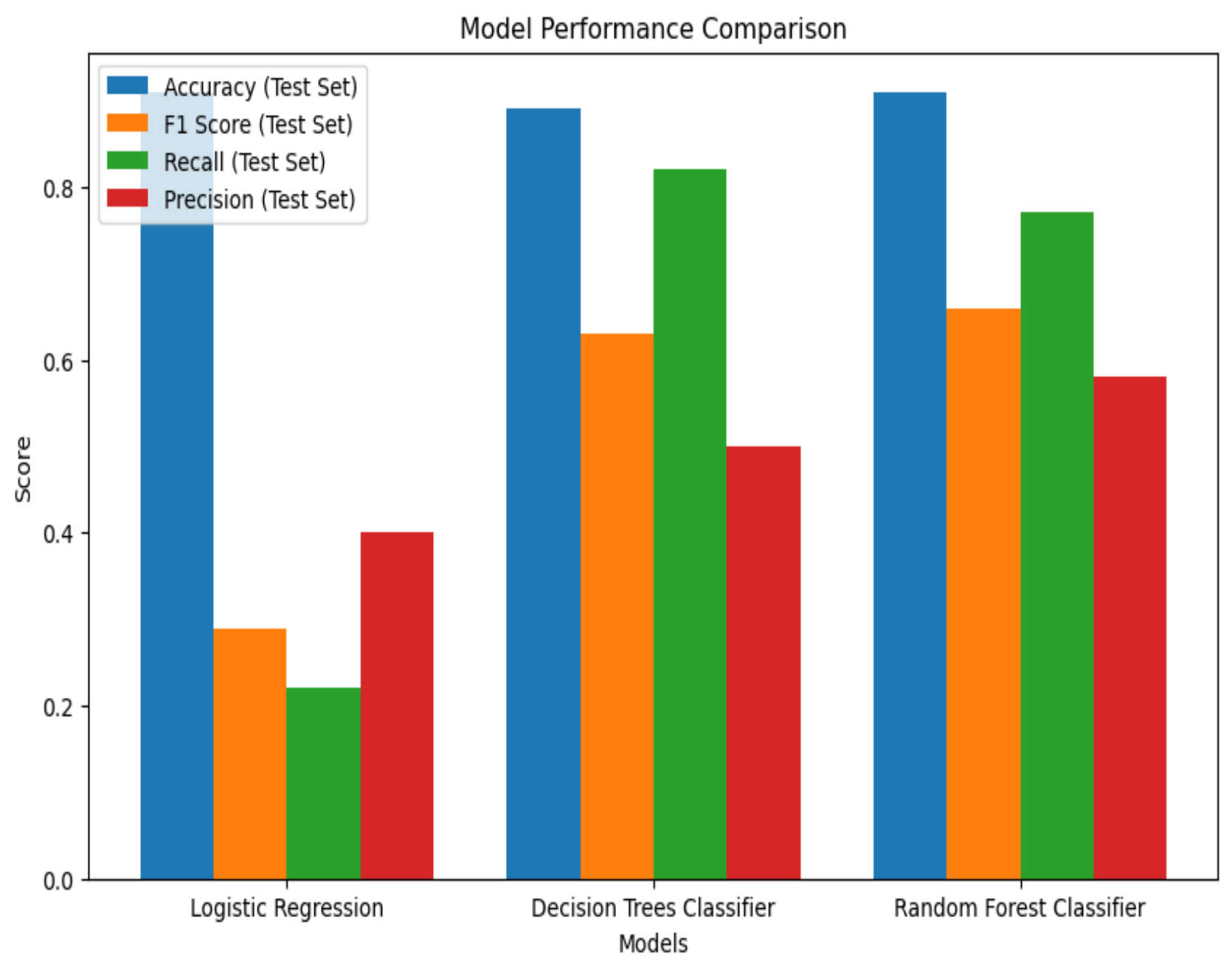
Random Forest Classifier performs better. This is explained by the high Accuracy, F1 Score , recall and Precision as highlighted on the metrics table.

Recall = 77%

F1 Score = 66%

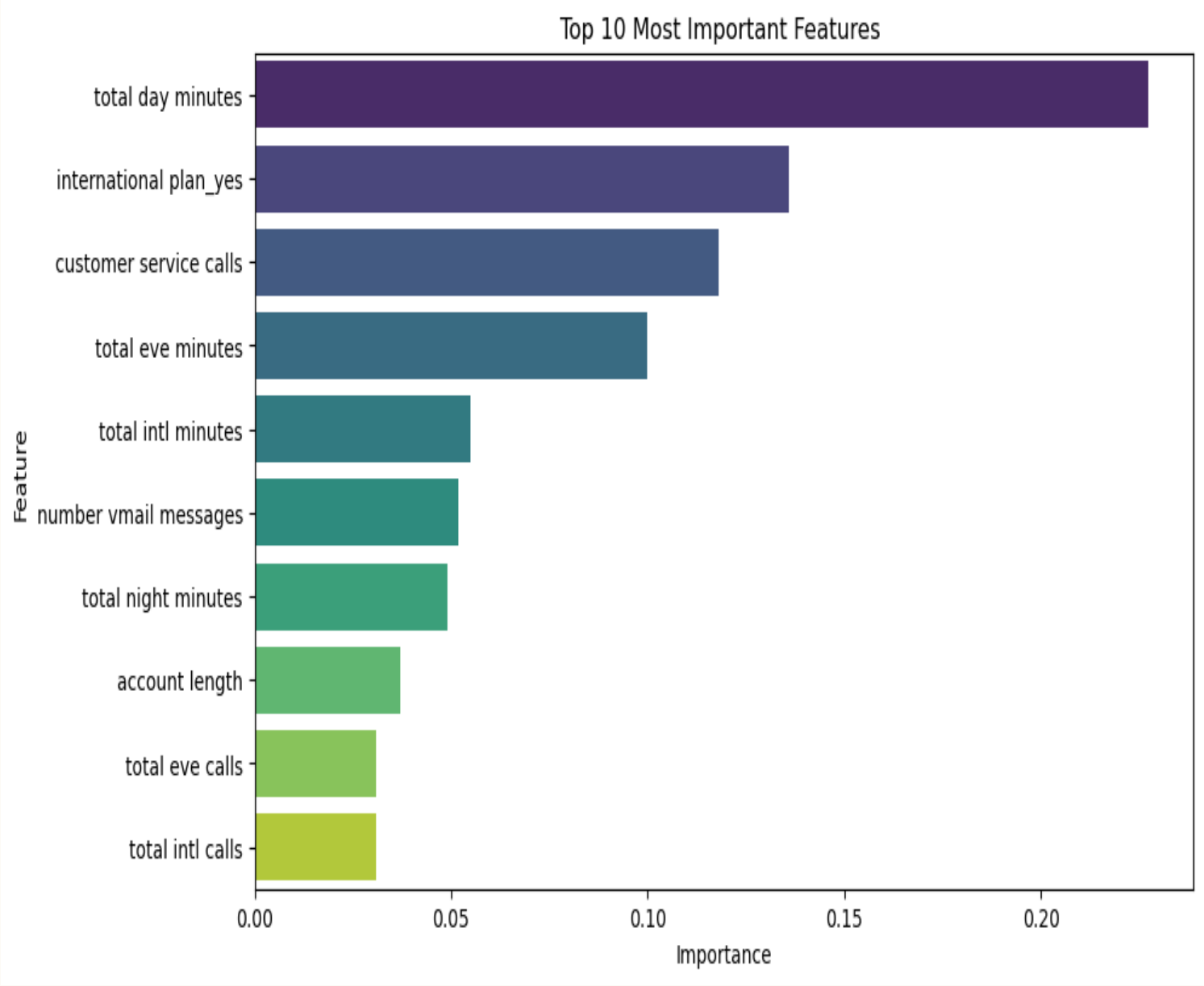
Accuracy = 91%

Precision = 58%



FEATURES INFLUENCING CUSTOMER CHURN

- Total duration of calls during the day and in the evening. Can be churn signals if reduce call duration.
- Total duration of international calls. Customers with high international usage may churn if they find cheaper alternatives.



RECOMMENDATIONS

- Syriatel adopt the Random Forest Classifier which has better evaluation metrics as the primary model for churn prediction.
- Syriatel should implement customer retention initiatives that target the key drivers of churn, particularly those related to call usage and pricing.
- The company should focus on improving customer service efficiency and satisfaction. Since the number of customer service calls is a significant predictor of churn.

NEXT STEPS

- To improve the effectiveness of this churn prediction project, the next step is to deploy the best-performing model into a real-world environment where it can be used to predict churn risk for active customers.
- Syriatel can then use these predictions to implement targeted retention strategies such as offering discounts, loyalty rewards, or improved customer support to customers identified as high-risk
- The company should focus on improving customer service efficiency and satisfaction. Since the number of customer service calls is a significant predictor of churn.

THANK YOU

QUESTIONS?

https://github.com/S-Mwaura/SyrialTel-Customer-Churn-Prediction-ML